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Experience

Ginkgo Bioworks: Biosecurity

Boston, Massachusetts

STAFF SOFTWARE ENGINEER

March 2021 - November 2025

Tech lead for a HIPAA-compliant pathogen surveillance platform - initially for COVID-19 in thousands of K-12 classrooms with lab partners across 18 states, then multipathogen biosurveillance across 11 international airports (126 origin countries) - a platform that processed 15M+ samples and underpinned \$733M in revenue.

- Led a multi-year platform evolution from single-pathogen COVID testing to multipathogen biosurveillance - refactoring live production systems while preserving historical data, existing visualizations, and CDC reporting pipelines.
- Designed the Results & Sample Refactor: decomposed a monolithic data entity into three domain objects (sample, procedure, result) with one-to-many relationships - enabling multi-panel testing without combinatorial data duplication.
- Led a team of ~11 as interim team lead through a period of rapid organizational change - running ceremonies, managing stakeholders, and keeping engineers focused on delivery; transitioned to a dedicated tech lead role once a manager was hired.
- DevOps: rebuilt CI/CD pipelines around Docker with GitHub Actions, codified infrastructure in Terraform (Auth0, AWS, Aptible, supporting services), migrated off legacy vendor systems, and achieved 8x build performance on self-hosted runners.
- Designed PHI removal from the platform as the business shifted from individual diagnostics to aggregate biosurveillance - balancing regulatory compliance, data retention, and system simplification.

🔧 Flask · SQLAlchemy · PostgreSQL · Terraform · GitHub Actions · Docker · Nix · AWS EC2/S3/CloudFront · Auth0 · Aptible · Grafana · New Relic · Heap

Oliver Wyman: Digital

Boston, Massachusetts

SENIOR SOFTWARE ENGINEER

October 2017 - February 2021

Software engineering across a broad range of projects and roles at a top-tier management consulting firm.

- DevOps: wrote idiomatic containers, stood up CI/CD into Kubernetes across Digital's project portfolio. Member of a Special Interest Group spreading DevOps best practices.
Synthesized an analytics stack to give consulting teams a platform for development, workflow scheduling, data operations, and execution.
- Webservices: primarily backends for analytics and database operations, interfaced with as REST and GraphQL API services.
Developed a system that continuously collects & collates data on all of Digital projects/assets - providing InfoSec auditability, surfacing lost items, and decreasing seek time for the sustaining teams.
- Data infrastructure: built tooling for consultants to use in ETL, workflow automation, schema management, and computational scaling.
Built out a migration-based schema framework for a project with a major grocery retailer, eliminating database down-time.
- Sustaining: led long-term operations for five projects in the mature phase of their lifecycle.

🔧 Python · Kubernetes & Kustomize · Jenkins Pipelines · Prefect · Node.js (Express & Vue) · Alembic · MS SQL Server · PostgreSQL · GraphQL

Lockheed Martin: Space Systems

Littleton, Colorado

SOFTWARE ENGINEER - ORION: NASA'S NEXT MANNED SPACECRAFT

September 2014 - September 2017

- Principal architect in moving Orion ground software from pen and paper to full test automation, empowering NASA requirement verification.
- Built a Python/LaTeX-based engine for creating NASA deliverables, cutting months of paperwork for engineers (per deliverable) to a day.
- Served as a subject matter expert on the flight software verification automation tiger team, sharing processes established while automating ground.

🔧 Python packaging · Pytest · LaTeX · Sphinx · Linux · Vim · Perforce · JIRA · Jenkins · Agile · Test-driven development · Elastic stack

Education

Colorado School of Mines B.S. ENGINEERING, MECHANICAL EMPHASIS – CS & ECONOMICS MINORS

December 2013

Capstone: NASA Lunabotics Competition: built a regolith mining robot and competed with international teams at the Kennedy Space Center.

Projects

Private Infrastructure Cloud

2015 - Present

Architected a multi-layer, GitOps-driven platform from bare-metal through immutable Kubernetes - hosting 24+ services. Archive-class file integrity.

🔧 OpenTofu/Terraform · Prometheus/Loki/Grafana · Flux CD · Nix · SOPS secrets · ZFS · TLS · Talos Linux · Proxmox · CoreDNS · Tailscale · OpenEBS